

AI Prompting Lab

Test Results & Response Report — All 13 Exercises

Assignment	AI Prompting Lab – 13 Hands-On Exercises
Focus	What response did each prompt produce? What was learned?
Tools Used	ChatGPT · Claude · Google AI Studio
Structure	Prompt Used → AI Response Received → Observation → Key Learning

Part 1 — How AI Knows Things

Exercise 1 Novice vs. Power User

■ ChatGPT

■ Prompt Used

"Simple: 'Which phone should I buy?' → then detailed version with budget \$300, photo usage, battery issue, size preference"

■ AI Response Received

- Simple prompt response: A list of popular phones (iPhone, Samsung, Pixel) with no specific recommendation — generic advice suitable for anyone.
- Detailed prompt response: ChatGPT recommended 3 specific budget phones (e.g., Samsung Galaxy A series, Pixel 6a), each with a one-line reason matching the given constraints.
- It also picked one final recommendation and explained why it suited the user's exact situation (battery life + camera + compact size).

■ Observation

- Simple prompt gave generic advice — true for anyone, useful to no one.
- Detailed prompt produced personalized recommendations based on budget, usage, and comfort.

■ Key Learning

- ✓ AI performs best when given context: budget, usage habits, and constraints.
- ✓ Without context, output becomes general and less useful to the actual user.

■ Prompt Used

"Step A: 'What is the difference between a CV and a cover letter?' → Step B: 'What is the current minimum notice period for contract changes in my country?'"

■ AI Response Received

- Step A response: Clear, accurate explanation — CV is a full work history; cover letter is a targeted introduction for a specific job. Confident and correct.
- Step B response: ChatGPT gave a general answer about notice periods but added a disclaimer: 'This may vary by country and may have changed — please verify with a legal source.'
- In some cases it gave a number confidently without disclaiming — showing the inconsistency.

■ Observation

- For general knowledge (CV vs cover letter), the response was accurate and reliable.
- For dynamic or location-specific info, AI sometimes admitted uncertainty — but not always.

■ Key Learning

- ✓ AI is reliable for stable, well-known knowledge.
- ✓ For real-world changing data (laws, news, prices), it may guess or disclaim — always verify.

■ Prompt Used

"Mode 1: 'Summarize Romeo and Juliet in 4 sentences' | Mode 2: 'What's the weather this weekend? Cite source.' | Mode 3: 'Research telecom bill — structured report with citations'"

■ AI Response Received

- Mode 1 (Memory): Instant response — Romeo and Juliet plot summarized in 4 clean sentences from pretrained knowledge. No search performed.
- Mode 2 (Web Search): ChatGPT searched the web, returned weekend weather forecast with a cited source and date — clearly labeled as real-time data.
- Mode 3 (Deep Research): Produced a structured report on telecom bill with 3 key points, a comparison table, open questions, and cited sources from official/government pages.

■ Observation

- Romeo & Juliet → instant memory-based response, no search needed.
- Weather → real-time web search performed, source cited.
- Telecom bill → full structured research report with citations generated.

■ Key Learning

- ✓ Trigger words like 'latest,' 'current,' 'cite sources' push AI into search or research mode.
- ✓ AI switches between memory, search, and structured reasoning based on how the prompt is written.

Part 2 — Talking to AI Well

■ Prompt Used

"I need help planning dinner. I have chicken, rice, onions, yogurt. Only 30 minutes. One person doesn't eat spicy. Give me 3 simple meal options, no commentary."

■ AI Response Received

- ChatGPT returned exactly 3 meal options: Chicken Rice Bowl, Yogurt-Marinated Grilled Chicken with Rice, and Onion-Chicken Stir Fry.
- Each option was labeled as non-spicy, prep time was under 30 minutes, and only the listed ingredients were used.
- No extra commentary, no irrelevant suggestions — output matched the brief exactly.

■ Observation

- Structured context (ingredients, time, dietary constraint) produced practical, ready-to-cook meal plans.
- Output became task-specific instead of generic recipe suggestions.

■ Key Learning

- ✓ Context is a quality multiplier — the more relevant detail you give, the better the output.
- ✓ Clear constraints (time, ingredients, restrictions) automatically remove irrelevant suggestions.

■ Prompt Used

"I am choosing between two job offers. Offer A: higher pay, long commute, less interesting. Offer B: lower pay, remote, steep learning curve. I value learning and family time. Think hard. Give me trade-offs, your pick, and when your answer flips."

■ AI Response Received

- ChatGPT listed 3 key trade-offs: financial stability vs. learning growth, commute time vs. family time, short-term comfort vs. long-term career value.
- It recommended Offer B, reasoning that the user's stated values (learning + family time) aligned more with remote work and a steeper learning curve.
- It also stated: 'My answer flips if the pay difference is more than 30% or if Offer A's commute is less than 20 minutes.'

■ Observation

- Job decision prompt produced a structured trade-off analysis, not just a simple yes/no.
- Output included pros/cons and conditional recommendations based on personal values.

■ Key Learning

- ✓ 'Think hard' improves depth and reasoning quality — AI slows down and considers multiple angles.
- ✓ AI performs structured decision analysis when explicitly asked for trade-offs and conditions.

■ Prompt Used

"Step A (biased): 'Don't you think mornings are obviously the best time to exercise?' → Step B (neutral): 'Compare morning vs evening exercise — strongest case for each. Don't tell me which to pick.'"

■ AI Response Received

- Step A response: ChatGPT agreed — listed benefits of morning exercise (metabolism boost, consistency, mental clarity) and barely mentioned downsides. Clearly biased toward the implied answer.
- Step B response: ChatGPT gave equal weight to both sides. Morning: builds routine, boosts focus. Evening: muscles are warmer, performance peaks, may improve sleep quality for some people.
- Step B also surfaced a point not previously considered: evening exercise can be better for strength training due to higher body temperature.

■ Observation

- Leading question produced biased agreement — AI listed reasons matching what was implied.
- 'Compare both sides' prompt produced a balanced, objective evaluation.

■ Key Learning

- ✓ Leading prompts produce biased answers — AI tends to agree with what you imply.
- ✓ Use neutral comparison prompts ('compare,' 'list both sides') to get honest, objective reasoning.

■ Prompt Used

"Round 1: '5 versions of a message asking a coworker to send a forgotten file — friendly, not annoyed. One line each.' → Round 2: feedback given → Round 3: expand winner"

■ AI Response Received

- Round 1: ChatGPT gave 5 different versions ranging from casual ('Hey! Any chance you can send that file?') to slightly formal. All were short, one-liner messages.
- Round 2 (after feedback 'I don't like option 3 — too casual. I like option 2 but make it warmer'): New 5 options were noticeably more refined in tone and matched the requested warmth.
- Round 3 (expand winner): Final message was 2–3 sentences, added a friendly closing line, and felt natural — significantly better than anything in Round 1.

■ Observation

- First output gave multiple options to choose from — useful range of approaches.
- Specific feedback in Round 2 improved tone and personalization noticeably.
- Final output after 3 rounds was much more refined and user-specific.

■ Key Learning

- ✓ AI output improves significantly through iteration — the loop is where real value comes from.
- ✓ Prompting is a conversation, not a one-time query. First answer is just the starting point.

Part 3 — Beyond Text

■ Prompt Used

"Uploaded a photo of a handwritten note / receipt. Prompt: 'Transcribe what it says. Mark unclear words as [unclear] with 2 best guesses.'"

■ AI Response Received

- Google AI Studio successfully extracted and transcribed all clearly written text from the image.
- Two words that were difficult to read were marked as [unclear] with two alternative guesses provided for each.
- The overall layout and structure of the original note was preserved in the output.

■ Observation

- AI successfully extracted text from the uploaded image.
- Unclear sections were appropriately flagged — AI did not silently guess wrong words.

■ Key Learning

- ✓ Multimodal AI can interpret non-text input such as photos, handwritten notes, and documents.
- ✓ Accuracy depends on image clarity — always review the flagged [unclear] sections yourself.

■ Prompt Used

"Build me a tip calculator. Goal: split a bill and add a tip. Input: total amount, tip %, number of people. Output: tip amount, grand total, each person's share. Show working version."

■ AI Response Received

- Claude generated a complete, working tip calculator UI in a single response — no errors, no missing features.
- The app had input fields for bill total, tip percentage, and number of people, with real-time output showing tip amount, grand total, and per-person share.
- When asked to iterate ('make buttons bigger, change to calm blue, add reset button'), Claude updated the existing app cleanly without breaking any functionality.
- Code was well-structured, readable, and more UI-focused compared to what ChatGPT generated for the same prompt.

■ Observation

- Successfully generated a working UI app using the Goal / Input / Output prompt structure.
- Code was more structured and UI-focused compared to ChatGPT's output for the same task.

■ Key Learning

- ✓ Claude performs strongly in frontend UI generation — clean, functional, and well-structured.
- ✓ Structured prompts using Goal / Input / Output format are critical for app-building tasks.

■ Prompt Used

"Round 1: Pasted 18 numbers, asked for median, average, and outliers — no mention of code.
Round 2: Same question but 'write and run code, show me the code you ran.'"

■ AI Response Received

- Round 1 response: ChatGPT gave a paragraph with calculated values — median and average presented confidently, but no code block appeared, meaning it computed mentally.
- Round 2 response: ChatGPT wrote a Python script, executed it, and showed both the code and the verified output: median = 65.5, average $\approx$ 61.6.
- The Round 1 answer happened to be correct in this case, but could have been wrong — Round 2 was verifiable and transparent.

■ Observation

- First response: quick mental calculation — no code shown, result not verifiable.
- Second response: Python code written, executed, and result verified — more reliable and transparent.

■ Key Learning

- ✓ Always request 'write and run code, show me the code' for any numerical calculations.
- ✓ No code block = AI probably guessed. A visible code block = verifiable and trustworthy result.

Part 4 — Working Safely & Choosing Tools

■ Prompt Used

""Imagine you have permission to reorganize 50 personal files. Write a SAFE step-by-step workflow so nothing is lost. Then list 3 things you should NEVER do.""

■ AI Response Received

- ChatGPT produced a 5-step safe workflow: (1) List all files and show current names, (2) Propose new names/folders for approval, (3) Wait for user confirmation before making any changes, (4) Make changes one batch at a time, (5) Confirm completion and show summary.
- The 3 things it said it should NEVER do: delete files permanently without confirmation, rename files in bulk without preview, or move files across drives without explicit permission.
- The response emphasized that no action should happen without a review and approval step first.

■ Observation

- AI produced a clear, step-by-step safe workflow for file handling tasks.
- It correctly emphasized permission and review stages before taking any action.

■ Key Learning

- ✓ AI should never execute file or system actions blindly — a plan must come first.
- ✓ Safe workflow order: Plan → Review → Approve → Act. Never skip the review step.

Exercise 12 Which Model When (Model Comparison)

■ ChatGPT + Claude

■ Prompt Used

"Same prompt in both tools: 'Write a 100-word LinkedIn post announcing I finished an AI prompting course. Friendly, not boastful.'"

■ AI Response Received

- ChatGPT response: Structured and polished — clear opening hook, logical flow, professional closing. Felt like a well-edited LinkedIn post ready to publish.
- Claude response: More natural and conversational in tone — felt warmer and more personal, but slightly less structured in its paragraph flow.
- For this specific task (professional LinkedIn writing), ChatGPT's output was more suitable due to better structure and formatting.
- Claude's output would have been preferable for a casual or creative writing task.

■ Observation

- ChatGPT: structured, analytical, professional — better for formal writing tasks.
- Claude: more natural and design-friendly — better for creative or conversational tasks.

■ Key Learning

- ✓ Different models have different strengths — no single model is best for all tasks.
- ✓ Always test the same prompt in two tools and use the better result for important work.

■ Prompt Used

"Pasted same 200-word draft into both tools: 'Score this 1–10 on clarity, structure, evidence, and what's missing. One sentence of reasoning per score. What single change raises the lowest score most?'"

■ AI Response Received

- ChatGPT scores: Clarity 8/10, Structure 7/10, Evidence 6/10, Missing: 'No real-world example to support the main claim.' Suggested fix: Add one specific case study.
- Claude scores: Clarity 7/10, Structure 8/10, Evidence 5/10, Missing: 'The argument lacks supporting data or statistics.' Suggested fix: Include at least one number or cited fact.
- Both agreed evidence was the weakest area — but Claude rated it lower and gave a more specific fix.
- ChatGPT did not flag the lack of statistics specifically; Claude caught this as the biggest gap.

■ Observation

- Both models gave slightly different evaluations of the same draft.
- Each identified different weaknesses — Claude caught a gap that ChatGPT did not emphasize.

■ Key Learning

- ✓ Cross-checking with two models improves accuracy — different models have different blind spots.
- ✓ Model disagreement highlights where your writing is genuinely weak. Use both critiques together.

Summary — What Each Prompt Produced

1	Novice vs Power User	ChatGPT	Personalized list	Generic → Specific with context
2	Knows vs Guesses	ChatGPT	Answer + disclaimer	Confident on facts, uncertain on dynamic info
3	Retrieval Modes	ChatGPT	3 different response types	Memory / Search / Research triggered by words
4	Context Is Everything	ChatGPT	Structured meal plan	3 practical options matching all constraints
5	Think Hard	ChatGPT	Trade-off analysis	Pros/cons + pick + condition when answer flips

6	Stop the Flattery	ChatGPT	Balanced comparison	Biased in Step A → Balanced in Step B
7	Brainstorm-Iterate	ChatGPT	Refined message	5 options → feedback → polished final version
8	Multimodal Input	Google AI Studio	Transcription	Text extracted, unclear parts flagged correctly
9	Build a Small App	Claude	Working UI app	Tip calculator built and iterated successfully
10	Data Analysis	ChatGPT	Python code + result	No-code guess vs. verified code-run result
11	Safe AI Permissions	ChatGPT	Safety workflow	5-step plan + 3 things AI should never do
12	Model Comparison	ChatGPT + Claude	Two different posts	ChatGPT better for structured professional writing
13	Models Checking Models	ChatGPT + Claude	Two critiques	Claude caught evidence gap ChatGPT missed

Overall Conclusion

Testing all 13 prompts across ChatGPT, Claude, and Google AI Studio showed one consistent pattern: **the quality of the AI's response is directly controlled by how well the prompt is written.** Every exercise where more context, structure, or specific instructions were given produced a noticeably better response than a vague or leading prompt.

The most important response differences observed were: (1) adding context turned generic answers into personalized ones, (2) asking for code produced verifiable results while not asking often produced silent guesses, (3) neutral comparison prompts gave honest evaluations while leading prompts gave flattery, and (4) iterating through multiple rounds consistently improved quality beyond what a single prompt could achieve.

Across all 13 exercises, the one rule that explained every result: **get the right context in — keep the wrong context out.**